Supplementary Material for "Thermodynamics of CO_2-loaded aqueous amines with ePC-SAFT. Part I: monoethanolamine calibration for phase equilibria, speciation, absorption heat, density and phase stability"

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S1 Model inputs and fixed-parameter provenance

Neutral MEA parameters are the 2B values of Baygi and Pahlavanzadeh, except the association volume described below [1, Table 2]. CO_2 segment count, diameter and dispersion energy are from Gross and Sadowski [2, Table 2]. Bicarbonate, carbonate and hydroxide sizes and dispersion energies follow Uyan et al.; their hydrogen-ion parameters are transferred here to hydronium [3, Table 4]. Ion segment count one and the diameter ties are modeling assumptions. Fixed Born diameters for bicarbonate, carbonate, hydronium and hydroxide, and the fixed water interactions with carbonate, hydronium and hydroxide, are model inputs without independent MEA-specific parameter measurements. All other interactions are zero by assumption except the fixed water interactions listed below. Tables S1 and S2 list their exact values; they are not attributed to the cited size/dispersion source. Water's solvation factor and ion-region permittivity follow Figiel et al.; unit solvation factors for the other components are assumptions [4, Table 2, Eqs. 5–10]. The ion packing and Debye–Hückel diameters follow the ties stated in the main text.

Table S1
Complete species inputs of the adopted parameter set. CO_2 association entries apply only to induced bonding with water. The water diameter law and the MEA association-volume conversion are specified in the text.

Species	Quantity (unit)	Value	Source
CO_2	m^{seg}	2.0729	Segments: [2, Table 2]; association: water fit (energy/2)
CO_2	σ (Å)	2.7852	Segments: [2, Table 2]; association: water fit (energy/2)
CO_2	ϵ/k_B (K)	169.21	Segments: [2, Table 2]; association: water fit (energy/2)
CO_2	ϵ^{AB}/k_B (K)	878.3249	Segments: [2, Table 2]; association: water fit (energy/2)
CO_2	κ^{AB}	0.037619728	Segments: [2, Table 2]; association: water fit (energy/2)
CO_2	z	0	Chemical charge
CO_2	f	1	Assumed
CO_2	d_V (L/mol)	1×10^{-8}	Assumed
CO_2	d_T (K)	1403	Assumed
MEA	m^{seg}	3.0353	Segments, energy: [1, Table 2]; association volume converted here
MEA	σ (Å)	3.0435	Segments, energy: [1, Table 2]; association volume converted here
MEA	ϵ/k_B (K)	277.174	Segments, energy: [1, Table 2]; association volume converted here

Species	Quantity (unit)	Value	Source
MEA	ϵ^{AB}/k_B (K)	2586.3	Segments, energy: [1, Table 2]; association volume converted here
MEA	κ^{AB}	0.036305439	Segments, energy: [1, Table 2]; association volume converted here
MEA	z	0	Chemical charge
MEA	f	1	Assumed
MEA	d_V (L/mol)	0.73098659	Permittivity fit
MEA	d_T (K)	1019.0803	Permittivity fit
H_2O	m^{seg}	1.2527538	Water fit
H_2O	σ (Å)	$\sigma_w(T)$	Water fit
H_2O	ϵ/k_B (K)	432.37881	Water fit
H_2O	ϵ^{AB}/k_B (K)	1756.6498	Water fit
H_2O	κ^{AB}	0.037619728	Water fit
H_2O	z	0	Chemical charge
H_2O	f	1.5	Water: [4, Table 2]; others: assumed
H_2O	d_V (L/mol)	0.3777	[5, Table 6]
H_2O	d_T (K)	1403	[5, Table 6]
$MEAH^+$	m^{seg}	1	Joint fit; segment count assumed
$MEAH^+$	σ (Å)	4.8558298	Joint fit; segment count assumed
$MEAH^+$	ϵ/k_B (K)	208.73557	Joint fit; segment count assumed
$MEAH^+$	z	+1	Chemical charge
$MEAH^+$	d^{Born} (Å)	4.8558298	Joint fit; segment count assumed
$MEAH^+$	f	1	Assumed
$MEACOO^-$	m^{seg}	1	Joint fit; segment count assumed
$MEACOO^-$	σ (Å)	5.3479914	Joint fit; segment count assumed
$MEACOO^-$	ϵ/k_B (K)	179.85787	Joint fit; segment count assumed
$MEACOO^-$	z	−1	Chemical charge
$MEACOO^-$	d^{Born} (Å)	5.3479914	Joint fit; segment count assumed
$MEACOO^-$	f	1	Assumed
HCO_3^-	m^{seg}	1	Size, energy: [3, Table 4]; Born: assumed
HCO_3^-	σ (Å)	2.9296	Size, energy: [3, Table 4]; Born: assumed
HCO_3^-	ϵ/k_B (K)	70	Size, energy: [3, Table 4]; Born: assumed
HCO_3^-	z	−1	Chemical charge
HCO_3^-	d^{Born} (Å)	3	Size, energy: [3, Table 4]; Born: assumed
HCO_3^-	f	1	Assumed
CO_3^{2-}	m^{seg}	1	Size, energy: [3, Table 4]; Born: assumed
CO_3^{2-}	σ (Å)	2.4422	Size, energy: [3, Table 4]; Born: assumed
CO_3^{2-}	ϵ/k_B (K)	249.26	Size, energy: [3, Table 4]; Born: assumed
CO_3^{2-}	z	−2	Chemical charge
CO_3^{2-}	d^{Born} (Å)	3	Size, energy: [3, Table 4]; Born: assumed
CO_3^{2-}	f	1	Assumed
H_3O^+	m^{seg}	1	Size, energy transferred: [3, Table 4]; Born: assumed
H_3O^+	σ (Å)	3.4654	Size, energy transferred: [3, Table 4]; Born: assumed
H_3O^+	ϵ/k_B (K)	500	Size, energy transferred: [3, Table 4]; Born: assumed
H_3O^+	z	+1	Chemical charge
H_3O^+	d^{Born} (Å)	1.218	Size, energy transferred: [3, Table 4]; Born: assumed
H_3O^+	f	1	Assumed
OH^-	m^{seg}	1	Size, energy: [3, Table 4]; Born: assumed
OH^-	σ (Å)	2.0177	Size, energy: [3, Table 4]; Born: assumed
OH^-	ϵ/k_B (K)	650	Size, energy: [3, Table 4]; Born: assumed
OH^-	z	−1	Chemical charge

Species	Quantity (unit)	Value	Source
OH^-	d^{Born} (Å)	3.0810769	Size, energy: [3, Table 4]; Born: assumed
OH^-	f	1	Assumed

Table S2
Binary interactions to eight significant figures. All unlisted interactions and slopes are zero; same-sign ion dispersion is excluded.

Pair	k_{ij}(313.15 K)	b_{ij} (K)	Source
CO_2–H_2O	−0.046416901	−35.665052	CO_2–water fit
MEA–H_2O	−0.088928359	−6.3627379	Joint fit
H_2O–$MEAH^+$	−0.35772543	96.174693	Joint fit
H_2O–$MEACOO^-$	−0.066121247	–	Joint fit
H_2O–$HCO_3{}^-$	0.76500002	–	Joint fit
$MEAH^+$–$MEACOO^-$	−1	–	Joint fit
H_2O–$CO_3{}^{2-}$	−0.25	–	Assumed
H_2O–H_3O^+	0.25	–	Assumed
H_2O–OH^-	−0.25	–	Assumed

The CO_2–water cross-association energy is 878.325 K, half the fitted water self-association energy, with the same association volume as water. The topology follows Schick et al. and Pabsch et al. [6, Table 1, Sect. 2.4.2][7]. The association strength here carries σ_{ij}^3, whereas Baygi and Pahlavanzadeh report their MEA association volume, 0.037470, for a d_{ij}^3 strength [1, Table 2]. The value used here, 0.036305439, is the constant fitted by least squares, with the ePC-SAFT library's fitter, to the vapor pressures and saturated-liquid densities of their own model at 21 temperatures from 293.15 to 393.15 K.

The main text reports the fits of the five water parameters to IAPWS-95 [8], the CO_2–water interaction to Kiepe pressures [9] and the MEA dielectric coefficients to permittivities. In the water fit, the two exponential terms in $\sigma_w(T) = \sigma_0 + a_1 \exp(b_1 T) + a_2 \exp(b_2 T)$, with $a_1 = 10.11$, $b_1 = -0.01775$, $a_2 = -1.417$ and $b_2 = -0.01146$, were fixed while $\sigma_0 = 2.7804326$ Å was fitted; diameters are in angstrom and temperature in kelvin.

In the permittivity fit, water coefficients, 0.3777 L/mol and 1403 K, follow Yang et al.; the CO_2 dielectric volume 10^{-8} L/mol and temperature 1403 K are assumptions [5, Table 6]. At the measurement temperature, two coefficient pairs give identical permittivities at any density. The larger-volume solution is used here; the permittivity data alone do not distinguish these solutions, so that selection is a model assumption. The 14 unloaded measurements at 298.15 K do not determine temperature or loading dependence; direct static-permittivity data over the reacting domain are missing from this calibration, so extrapolation remains a model assumption.

S2 Objective scales and weights

Table S3 gives the scales, weights and contributions to the joint objective of main-text Eq. 26. Pressure residuals are log ratios with unit scale; other residuals divide a difference by a scale in the observable's units. Species fractions and amount ratios use the assigned scale $0.1y^{\mathrm{obs}} + 0.001$. The fitted heat-capacity scale is $0.02C_P^{\mathrm{obs}}/n_{\mathrm{feed}}$ in J/(mol feed K), not a mass-specific heat capacity.

Vinjarapu heat scales use the printed uncertainty converted to J/mol CO_2 [10, Table 4]. The species relative allowances and floors, log-pressure allowance, carbon-free-density allowance, Cai allowances and group weights are assigned choices without a statistical derivation. Cai's reported temperature resolution and maximum composition error differ from the fitting allowances [11, Tables 2–3]. The near-unit residual variance factor is not a statistical justification for these weights. Scales inherited from source accuracy statements and assigned residual scales remain distinct.

Table S3
Assigned objective scales and weights. Rows with identical group and quantity are combined by range; the weight and scale are distinct. Origins and unrecorded rationales are described in the text.

Group; quantity	n	Scale range	Weight range	Cost
Pressure, species and heat; CO_2 pressure	47	1	11.1111	5.96255
Pressure, species and heat; mole fraction	94	0.0014–0.01154	1	16.4867
Pressure, species and heat; integral heat	8	3000–6000	1	1.38648
Additional species; mole fraction	7	0.00244–0.01109	1	2.40556
Jakobsen ratios; amount ratio	6	0.00172793–0.0548869	1	12.7457
CO_2–water; pressure	39	1	0.025641	0.000679019
CO_2-free density; density	5	0.5	10	13.1168
MEA–water; temperature	25	1	1	39.8239
MEA–water; mole fraction	25	0.01	1	51.8074
Loaded density; density	47	1.79111–2.41635	0.0212766	0.378743
Heat capacity; equilibrium heat capacity	4	1.59309–1.76153	1	6.64835

S3 Source conventions and density scales

Amundsen's density and loading inputs are $2\,\mathrm{kg\,m^{-3}}$ and 2 % of loading for the fitted loaded series, with unresolved high-temperature/loading applicability of the density estimate [12, pp. 3099–3100]. Hartono's inputs are $0.02\,\mathrm{kg\,m^{-3}}$ and 0.01 mol/mol loading; the density input is the printed expanded uncertainty with coverage factor two and is used without division by that factor [13, Table 3, p. 1810]. The calculations adopt Table 3's printed loading α. The accompanying printed CO_2 mole fractions imply different loadings under the stated solvent composition; that source inconsistency remains unresolved. The density comparisons and conditional parameter errors therefore depend on the adopted printed-loading convention. The secant construction in main-text Eq. 25 combines these with loading uncertainty. These scales therefore do not uniformly represent standard uncertainties.

All density observations, including the excluded Amundsen loading-0.4 series, remain in the electronic data with source pressure, composition and calculated value. The source comparison does not identify one experiment as correct. For Jakobsen NMR calculations, dilution changes the solvent composition, and the reported NMR loading is used instead of the maximum loading [14, Sample preparation, Tables A1–A2]. The carbonate-share comparison differs from the fitted species-to-total-amine ratios.

S4 Caloric functions and reproduction values

Neutral caloric functions use NIST-JANAF for CO_2, the MEA heat-capacity function tabulated by Zhang et al., and an ideal-gas water fit to IAPWS-95 [15, 16, 8]. The ionic functions are completed from the reaction laws and their infinite-dilution reference corrections. A charge-proportional reference freedom cancels in electroneutral phases; no independent ionic heat-capacity functions are fitted. The reaction enthalpy is included once through these completed standard states. Table S4 expands the main-text reproduction states with the nonzero energy terms and liquid fractions.

The primary common-row Kim comparison includes calculated intervals whose final loading is at most 0.60 mol CO_2/mol total MEA. It includes the two inconsistent printed observations at 393.15 K, run 2, with final loadings 0.078 and 0.445 mol/mol. The source-consistent alternative omits those two observations, and a further diagnostic also omits the first interval of each run at that temperature. Table S5 reports these alternatives without choosing a parameter set from them. All use the same imposed-pressure approximation and finite-addition definition as the main comparison.

Table S4
Reduced residual molar Helmholtz contributions $A_i/(nRT)$ and true liquid species fractions at the three main-text reproduction states. Zero terms are omitted.

Contribution or species	CO_2–water	MEA–water	Loaded MEA
Association	−2.914216572	−3.283777415	−2.548146614
Born	0	0	−6.794645267
Debye–Hückel	0	0	−0.03722854572
Dispersion	−10.8483615	−12.82081511	−10.56887359
Hard chain	4.893585711	7.127203367	4.784395775
Total residual	−8.868992357	−8.977389156	−15.16449824
$x_{HCO_3^-}$	0	0	0.001204998998
x_{MEACOO^-}	0	0	0.0324533626
x_{CO_2}	0.00036	0	$3.760071054 \times 10^{-7}$
$x_{CO_3^{2-}}$	0	0	$7.270063478 \times 10^{-6}$
$x_{H_3O^+}$	0	0	$5.901616356 \times 10^{-11}$
x_{OH^-}	0	0	$8.380416379 \times 10^{-8}$
x_{MEA}	0	0.9264	0.04609367749
x_{MEAH^+}	0	0	0.03367298547
x_{H_2O}	0.99964	0.0736	0.8865672455

Table S5
Kim finite-interval heat comparisons at 393.15 K. RMSE is kJ/mol absorbed CO_2; AARD is percent. All alternatives use the same F2 parameters.

Rows	n	RMSE	AARD
All printed	20	31.7735	57.6357
Without inconsistent rows	18	28.6239	21.8917
Also without first intervals	16	19.3967	19.3875

S5 Conditional parameter uncertainty

The ten interior parameters have local covariance from main-text Eq. 28; the ion–ion interaction at its active lower bound remains fixed. The weighted Jacobian has rank ten, and the variance factor is 1.01523840323. Table S6 gives values, bounds and conditional standard errors, and Table S7 gives the correlations in the same interior-parameter order. Correlated errors within one experimental source, model discrepancy, source-law uncertainty and future measurement noise are not included.

Table S6
Jointly fitted parameters, bounds and conditional standard errors. A dash marks the active bound. Born diameters of the two amine ions equal their segment diameters.

Parameter	Value	Lower	Upper	SE	Source
$MEACOO^-$–water k_{ij}	−0.066121247	−0.5	0.5	0.5092	Joint fit
$MEAH^+$–water k_{ij}	−0.35772543	−0.5	0.5	0.582	Joint fit
$HCO_3{}^-$–water k_{ij}	0.76500002	−0.5	1	2.965	Joint fit
$MEACOO^-$–$MEAH^+$ k_{ij}	−1	−1	1	–	Joint fit
$MEAH^+$–water k_{ij} temperature slope	96.174693	−1000	1000	32.26	Joint fit
$MEAH^+$ segment diameter	4.8558298	2	5.8	0.9514	Joint fit
$MEACOO^-$ segment diameter	5.3479914	2	5.8	0.8393	Joint fit
MEA–water k_{ij}	−0.088928359	−0.5	0.5	0.002234	Joint fit
MEA–water k_{ij} temperature slope	−6.3627379	−1000	1000	2.885	Joint fit
$MEAH^+$ dispersion energy/k	208.73557	50	950	80.3	Joint fit
$MEACOO^-$ dispersion energy/k	179.85787	50	1000	59.45	Joint fit

Table S7
Conditional correlations of the ten interior parameters, ordered as the fitted-parameter table with its active-bound row omitted.

Parameter	1	2	3	4	5	6	7	8	9	10
1	1									
2	−0.988	1								
3	0.96	−0.974	1							
4	−0.326	0.206	−0.199	1						
5	−0.101	0.231	−0.319	−0.702	1					
6	0.0841	−0.213	0.303	0.703	−0.998	1				
7	−0.864	0.826	−0.794	0.535	−0.139	0.153	1			
8	−0.767	0.73	−0.702	0.548	−0.141	0.153	0.92	1		
9	0.398	−0.461	0.633	0.208	−0.757	0.752	−0.22	−0.188	1	
10	−0.52	0.573	−0.731	−0.0455	0.595	−0.584	0.38	0.332	−0.941	1

The pressure intervals propagate the full correlated covariance rather than independent marginal parameter errors (fig. S1). They use the local Gaussian approximation and may extend below zero pressure. The inward scans in Table S8 fix the bound interaction at successive fractions of its feasible interval and refit the remaining parameters. They converge as numerical diagnostic calculations but have no renewed stability qualification and do not locate confidence limits or a global optimum.

S6 Phase-search conditions and split interpretation

The liquid grid contains 1800 states at 15, 25, 35 and 45 wt% MEA, 313.15–393.15 K in 10 K steps, and loading 10^{-5} plus 0.025–0.6 mol/mol in 0.025 steps, at both pressure ratios stated in the main text. Another 108 states use the liquid–vapor-search feed at those two ratios. The separate 576-state liquid check uses 20, 30 and 40 wt%, 318.15–388.15 K and loadings 0.0125–0.5875, at 1.5 times bubble pressure.

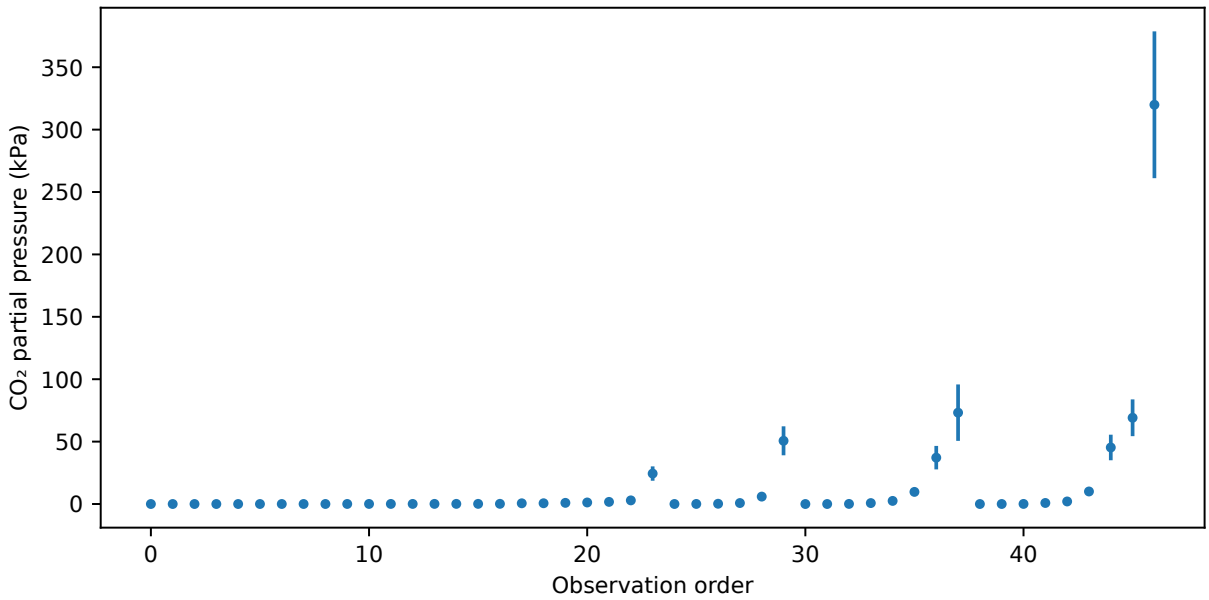


Figure S1: All fitted pressure states with the adopted calculated pressure (points) and correlated conditional intervals (bars). The interval is ±1.96 conditional standard errors with the active-bound parameter fixed; it is not an interval for a new observation. Values are ordered by observation identity.

Table S8
Inward scans of the ion–ion interaction, with the other parameters refitted. These are diagnostic fits without renewed stability qualification.

Inward step (percent)	Fixed interaction	C	Change in C
1	−0.98	150.891432	0.1285292
5	−0.9	151.434669	0.6717658
10	−0.8	152.161581	1.398678

The liquid–vapor search contains six temperatures, 313.15, 318.15, 333.15, 353.15, 373.15 and 393.15 K, and nine loadings, 10^{-5}, 0.05, 0.1, 0.2, 0.256, 0.3, 0.4, 0.5 and 0.6 mol/mol. Each uses three pressure fractions, 0.50, 0.80 and 0.95, of a fixed reference bubble pressure from the comparison calculation. Every compared parameter set is evaluated at these same physical states. The feed is 9.7292821 mol MEA, 76.9751925 mol water and 9.7292821α mol CO_2. Thus extensive split-energy changes divided by RT are in mol on this feed basis.

Charge-neutral liquid starts include a uniform composition and near-pure neutral-species compositions; vapor starts contain only the three neutral species. The optimized search and the evaluations at split compositions resolved for other MEA parameter sets are reported separately. Only a negative tangent-plane distance followed by a material-conserving finite transfer beyond the numerical energy-error estimate counts as a resolved descent. Search completion, residual convergence and positive pressure–density derivatives are numerical checks rather than physical measurements. The raw state results identify incomplete or unavailable cases explicitly.

The split-energy figure evaluates the same phase compositions under two parameter sets and closes density at the imposed pressure for each. It does not relax the frozen F2 phases to equilibrium, nor identify a microscopic interaction uniquely. The isolated segment-diameter change leaves the other diameter definitions unchanged and is outside F2's fitted diameter ties. The alternative sufficient interaction change also lacks fitted-property evidence. Neither probe can be compared with F2's covariance as if it were an uncertainty interval for liquid splitting.

S7 Shared inputs for Part II

The shared coefficient table in the main text lists the adopted F2 values. Part II also keeps the shared component charges, segment counts, association topology, electrostatic model choices, standard states and R1–R3 reaction definitions fixed. The water–bicarbonate interaction is the MEA-fitted value, whereas the carbonate–water interaction is an assumed fixed value. The weakly determined bicarbonate interaction and the dielectric and source-law extrapolations limit the strength of a transfer claim before amine-specific comparisons.

References

[1] S. Fakouri Baygi, H. Pahlavanzadeh, Application of the perturbed chain-SAFT equation of state for modeling CO_2 solubility in aqueous monoethanolamine solutions, Chemical Engineering Research and Design 93 (2015) 789–799.

[2] J. Gross, G. Sadowski, Perturbed-Chain SAFT: An Equation of State Based on a Perturbation Theory for Chain Molecules, Industrial and Engineering Chemistry Research 40 (2001) 1244–1260.

[3] M. Uyan, G. Sieder, T. Ingram, C. Held, Predicting CO_2 solubility in aqueous N-methyldiethanolamine solutions with ePC-SAFT, Fluid Phase Equilibria 393 (2015) 91–100.

[4] P. Figiel, G. Yu, C. Held, Predicting Thermodynamic Properties of Ions in Single Solvents and in Mixed Solvents Using a Modified Born Term within the ePC-SAFT Framework, Industrial and Engineering Chemistry Research 64 (2025) 9406–9418.

[5] F. Yang, G. M. Kontogeorgis, J. C. de Hemptinne, Composition-dependence of relative static permittivity in ePPC-SAFT for mixed-solvent alkali halides, Fluid Phase Equilibria 583 (2024) 114103.

[6] D. Schick, L. Bierhaus, A. Strangmann, P. Figiel, G. Sadowski, C. Held, Predicting CO_2 solubility in aqueous and organic electrolyte solutions with ePC-SAFT advanced, Fluid Phase Equilibria 567 (2023) 113714.

[7] D. Pabsch, C. Held, G. Sadowski, Modeling the CO_2 Solubility in Aqueous Electrolyte Solutions Using ePC-SAFT, Journal of Chemical and Engineering Data 65 (2020) 5768–5777.

[8] International Association For The Properties Of Water And Steam, Revised Release on the IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and

Scientific Use, Technical Report R6-95(2018), International Association for the Properties of Water and Steam, 2018.

[9] J. Kiepe, S. Horstmann, K. Fischer, J. Gmehling, Experimental Determination and Prediction of Gas Solubility Data for CO_2 + H_2O Mixtures Containing NaCl or KCl at Temperatures between 313 and 393 K and Pressures up to 10 MPa, Industrial & Engineering Chemistry Research 41 (2002) 4393–4398.

[10] S. H. B. Vinjarapu, T. Regueira, R. Neerup, N. von Solms, P. L. Fosbøl, Heat of Absorption of CO_2 in 30 Wt% MEA with Monoethyleneglycol and Urea as Vapour Reduction Additives, Energy 293 (2024) 130609.

[11] Z. Cai, R. Xie, Z. Wu, Binary Isobaric Vapor-Liquid Equilibria of Ethanolamines + Water, Journal of Chemical & Engineering Data 41 (1996) 1101–1103.

[12] T. G. Amundsen, L. E. Øi, D. A. Eimer, Density and Viscosity of Monoethanolamine + Water + Carbon Dioxide from (25 to 80) °C, Journal of Chemical & Engineering Data 54 (2009) 3096–3100.

[13] A. Hartono, E. O. Mba, H. F. Svendsen, Physical Properties of Partially CO_2 Loaded Aqueous Monoethanolamine (MEA), Journal of Chemical & Engineering Data 59 (2014) 1808–1816.

[14] J. P. Jakobsen, J. Krane, H. F. Svendsen, Liquid-Phase Composition Determination in CO_2-H_2O-Alkanolamine Systems: An NMR Study, Industrial and Engineering Chemistry Research 44 (2005) 9894–9903.

[15] M. W. Chase, NIST-JANAF Thermochemical Tables, 4th Edition, American Institute of Physics, 1998.

[16] Y. Zhang, H. Que, C. C. Chen, Thermodynamic modeling for CO_2 absorption in aqueous MEA solution with electrolyte NRTL model, Fluid Phase Equilibria 311 (2011) 67–75.